INCH-POUND

MIL-C-20354C (AR)
AMENDMENT 2
23 June 1994
SUPERSEDING
AMEDNDMENT 1
May 7 1993

MILITARY SPECIFICATION

CARTRIDGE, 105MM, ILLUMINATING
M314A3, LOADING, ASSEMBLING AND PACKAGING

This amendment forms a part of MIL-C-20354C (AR), dated 19 May 1988, and is approved for use by the U.S. Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

* 2.1.2, Other Government documents, drawings, and publications. Delete drawing 7549072 in its entirety and replace with:

"12934901	-	Packing and marking for container ammo metal, PA117, for 105mm Howitzer ctg. packed in PA55 or M105A3
19-48-4231/ 45-20PM1006	-	Unitization procedures for ammunitions components packed in a cylindrical metal or plastic container on a 4-way entry metal pallet"

* Change "M105A2" to "M105A3" wherever it appears.

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3.2.3, Burning time - candle power of the illuminant assembly (dwg. 9206749).: Delete in its entirety and replace with:

"3.2.3 Luminosity and burning time. The average luminosity of the canister assemblies shall be not less than 525,000 candlepower. The burning time of the canister assemblies shall be not less than 60 seconds. Determination shall be as specified in 4.4.3.5 and 4.5.3.

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TABLE 1, First Article Inspection: Add the following new inspections to Iluminating Canister:

"Illuminant composition test	4 ounces	3.2.1	4.5.2.2/ 4.5.2.3
Luminosity and burning time	5 pair	3.2.3	4.4.3.5/ 4.5.3"

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TABLE 1, First Article Inspection: First Fire Composition: Delete in its entirety and replace with:

"First Fire Composition First Fire Composition test	4 ounces	3.2.1	4.5.2.1/4.5.2.3"

* Add the following new first article tests:

"Container, Ammo. Fiber, M105A3, for 105mm Howitzer Ammunition (Dwg. 7549073) Examination for defects	5	3.1	4.2.2.11/ 4.2.2.12
Packing and Marking for Container Ammo Metal, PA117, for 105mm Howitzer Ctg. Packed in PA55 or M105A3 (Dwg. 12934901) Examination for defects	5	3.1	4.2.2.13

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Palletized Projectile, M314A3; Packed in accordance with drawing 19-48-4231/ 45-20PM1006			
Examination for defects	1 pallet	-	4.2.2.14"

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4.2.2.7 Add the following Major Defect:

Major				
108	Plug disassembly torque, max. (if applicable)	0.40%	3.1	gage

PAGE 21

* 4.2.2.11. Delete in its entirety and replace with new paragraph 4.2.2.11 (included).

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* 4.2.2.12. Delete in its entirety and replace with new paragraph 4.2.2.12 (included).

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* Add new paragraphs 4.2.2.14 and 4.2.2.15 (included).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.2.2.11	Unsealed Fiber Container	1 OF 1	7549073
			NEXT HIGHER ASSEMBLY: 7549073 (sealed)

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Packing component missing, damaged, or improperly assembled	0.40%	3.1	Visual
102	Cartridge cannot be inserted or removed by hand	0.40%	3.1	Manual
103	Container covers cannot be removed by hand	0.40%	3.1	Manual
104	Insufficient fillers for tight pack	0.40%	3.1	Gage
Minor				
201	Evidence of poor workmanship	0.65%	3.6	Visual/Manual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

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AMENDMENT 2

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.2.2.12	Sealed fiber container	1 OF 1	7549073
			NEXT HIGHER ASSEMBLY: 12934901

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Sealing strip incomplete	0.40%	3.1	Visual
102	Sealing strip wrinkled	0.40%	3.1	Visual
103	Sealing strip material improper	0.40%	3.1	Visual/Certification
104	Sealing strip damaged	0.40%	3.1	Visual
105	Interior fiber moisture content excessive at time of pack	0.40%	3.1	Gage
106	Component missing	0.40%	3.1	Visual
107	Marking misleading, missing, incorrect, or illegible	0.40%	3.1	Visual
108	Tight pack	0.40%	3.1	Manual/Audible
Minor				
201	Tear tab length below minimum	0.65%	3.1	Gage/SME
202	Component damaged	0.65%	3.1	Visual
203	Evidence of poor workmanship	0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.2.2.14	Sealed metal container	1 OF 1	12934901

NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Container damaged exposing contents	0.40%	3.1	Visual
102	Metallic seal missing or improperly secured	0.40%	3.1	Visual/Manual
103	DoD symbol missing or incorrect	0.40%	3.1	Visual
104	Air leak test	100%	3.1	4.5.4
105	Screw not properly replaced and torqued after test	0.40%	3.1	Visual/Gage
106	Contents loose inside metal container	0.40%	3.1	Manual/Audible
107	Interior packing material moisture content excessive at time of pack	0.40%	3.1	Gage
108	Cloth strap incorrect width or length	0.40%	3.1	SME/Gage
109	Cloth strap assembled incorrectly	0.40%	3.1	Visual
110	Cloth strap missing	0.40%	3.1	Visual
111	Marking incorrect, missing, or unidentifiable	0.40%	3.1	Visual
Minor				
201	Evidence of poor workmanship	0.65%	3.6	Manual/Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.2.2.15	Palletized Projectile, M314A3; packed in accordance with drawing 19-48-4231/45-20PM1006	1 OF 1	NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Strapping missing	Note 1	3.6	Visual
102	Strapping loose	Note 1	3.6	Visual/Manual
103	Incorrect number of containers on pallet (except for light load)	Note 1	3.6	Visual
104	Containers not all positioned in the same direction	0.40%, Note 2	3.6	Visual
105	Containers not properly interlocked	0.40%, Note 2	3.6	Visual
106	Marking missing, incorrect, or illegible	0.40%, Note 2	3.6	Visual
Minor				
201	Evidence of poor workmanship	0.65%, Note 2	3.6	Visual/Manual

NOTES:

Note 1 - Sample size to be the smallest number of pallets that contain a quantity of projectiles that reflects an AQL of 0.40%.

Note 2 - If inspection requires pallets to be opened, the entire quantity of projectiles on that pallet may, at the Government inspector's option, be applied to the sample quantity, provided that the pallet is selected randomly.

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4.4.3.5 Static test of illuminant canister assembly. Delete in its entirety and replace with:

"4.4.3.5 Luminosity and burning time (see 3.2.3 and drawing 9206749). The test shall be performed as specified in 4.5.3, and in accordance with Dwg. 9206749. A sample of two canister assemblies shall be selected in accordance with the sampling plan shown in 4.4.3.5. One canister assembly shall be tested for static luminosity and burning time. If the luminosity is less than 525,000 candlepower, the second candle shall also be tested. If the resulting average of the combined sample is less than 525,000 candlepower (luminosity), inspection lots represented by the samples shall be rejected (see 4.4.3.5.1.7). No one sample can exhibit a burning time of less than 60 seconds. The candlepower integration time shall be from the best continuous 55 seconds of burn time."

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4.4.3.5.1: Delete in its entirety and replace with:

"4.4.3.5.1 Sampling plan for luminosity and burning time. An inspection lot for the canister assembly shall be eighty-five candles produced consecutively from one mixing and pressing operation, and will be designated as one mix/batch for identification purposes (see 6.8). Smaller lot sizes will be accepted when dictated by interruption or shutdown of operations. The illuminant, as it is mixed, shall be dispensed into containers which shall be numbered and used in order."

4.4.3.5.1.1: Delete in tis entirety and replace with:

"4.4.3.5.1.1 Level one sampling. A sample of two candles from each inspection lot shall be selected at random. The canister assembly shall be tested for compliance with the requirements specified in paragraph 3.2.3. This sampling frequency shall be continued until fifteen consecutive samples of two candles comply with the requirements of paragraph 3.2.3."

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4.4.3.5.1.2 Delete in its entirety and replace with:

"4.4.3.5.1.2 Level two sampling. After successful completion of level one sampling, the sampling frequency may be reduced to two samples from each group of six inspection lots. The canister assemblies shall be tested for compliance with the requirements specified in paragraph 3.2.3. This sampling frequency shall be continued until fifteen consecutive samples of two candles comply with the requirements of paragraph 3.2.3. In the event that one lot fails to comply with the requirements, the sampling shall revert to level one sampling."

4.4.3.5.1.3: Delete in its entirety and replace with:

"4.5.3.5.1.3 Level three sampling. After successful completion of level two sampling, the sampling frequenecy may be reduced to two samples from each group of fifteen inspection lots. The canister assemblies shall be tested for compliance with the requirements specified in paragraph 3.2.3. This sampling frequency shall be continued until fifteen consecutive samples of two candles comply with the requirements of paragraph 3.2.3. In the event that one lot fails to comply with the requirements, the sampling shall revert to level two sampling."

4.4.3.5.1.4: Delete in its entirety and replace with:

"4.4.3.5.1.4 Level four sampling. After successful completion of level three sampling, the sampling frequency may be reduced to two samples from each shift (not to exceed ten hours) of production. The canister assemblies shall be tested for compliance with the requirements specified in paragraph 3.2.3. In the event that one lot fails to comply with the requirements, the sampling shall revert to level three sampling."

4.4.3.5.1.5: Delete in its entirety and replace with:

"4.4.3.5.1.5. In the event of any production stoppape of less than two weeks, the sampling plan shall revert to level one, with only seven consecutive samples required to advance to each reduced level until the level of sampling in use when production is reached."

4.4.3.5.1.6: Delete in its entirety and replace with:

"4.4.3.5.1.6. If the mixing operation has not been shut down, but merely diverted to another item, and has not experienced any cause for rejection per the sampling plan of that item(s), the sampling level shall remain unchanged, and fifteen consecutive successful samples of two candles shall be required to advance to the next level, if applicable."

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4.4.3.5.1. Sampling plan for luminosity and burning time: Add new subparagraph 4.4.3.5.1.7:

"4.4.3.5.1.7 In the event of failure of any sample to comply with the requirements, each individual inspection lot within the mix/batch must be tested. Failure of an inspection lot to comply with the requirements shall result in rejection of that particular inspection lot."

PAGE 27

4.5.2.1. First Fire Composition: Change "Equal parts by volume of carbon tetrachloride and anhydrous methanol" to "Equal parts by volume of dimethylformamide and anhydrous methanol".

PAGE 28

4.5.3. Delete in its entirety and replace with:

"4.5.3 Luminosity and burning time. The discs shall be removed from the candle prior to testing. In calculating the candle power of any one illuminant assembly, the best continuous 55 seconds of illuminometer readings are to be used."

* Add new paragraph 4.5.4 to follow paragraph 4.5.3:

"4.5.4 Air pressure test. The packed metal container assemblies shall be subjected to the internal air pressure test as specified on drawing 12934901. Observation shall be made for a pressure drop exceeding the requirement for the specified time. The cartridge packed in a defective container may be removed and packed in a replacement container to be tested."

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* 5.1.1. Change the first sentence to read "Each cartridge shall be packaged in accordance with drawing 7549073."

* 5.2.1. Delete the first sentence and replace with "Each packaged fiber container (Dwg. 7549073) shall be packed in a PA117 metal container and cushion assembly in accordance with drawing 12934901.".

* Add new paragraph 5.2.2:

"5.2.2 Level A - Alternative. Only with the specific approval of the contracting officer, the cartridge in fiber container shall be packed in a wooden box in accordance with dwg. 7549072."

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* 5.3. Delete in its entirety and replace with the following:

"5.3 Marking.

5.3.1 Level A. The fiber container shall be marked in accordance with drawing 7549073.

5.3.2 Level A. The metal container outer pack shall be marked in accordance with drawing 12934901 and MIL-STD-129.

5.3.3 Level A (wood box). The wood box shall be marked in accordance with dwg. 7549073 and MIL-STD-129."

* Add new paragraph 5.4 to follow 5.3:

"5.4 Unitiziation.

5.4.1 Level A. The packed metal container shall be unitized in accordance with drawing 19-48-4231/45-20PM1006."

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